

JARUD QI, China

WMO: 540260

Lat: 44.5473N

Lon: 120.8988E

Elev: 266

StdP: 98.17

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.8	(c) -19.0	(d) -31.6	(e) 0.2	(f) -17.7	(g) -29.7	(h) 0.2	(i) -16.0	(j) 8.1	(k) -9.5	(l) 7.3	(m) -11.0	(n) 2.3	(o) 290	(p) 0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.6	(c) 34.2	(d) 20.0	(e) 32.5	(f) 19.7	(g) 30.9	(h) 19.4	(i) 23.9	(j) 29.2	(k) 22.9	(l) 28.2	(m) 21.9	(n) 27.3	(o) 3.1	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.4	17.6	26.8	21.2	16.4	26.1	20.1	15.3	25.2	73.6	29.3	69.5	28.2	65.8	27.4	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.3	(b) 6.4	(c) 5.5	(e) -22.4	(f) 37.6	(g) 2.5	(h) 2.1	(i) -24.2	(j) 39.1	(k) -25.7	(l) 40.3	(m) -27.1	(n) 41.5	(o) -28.9	(p) 43.0		
(a) 7.3	(b) 6.4	(c) 5.5	(e) -23.0	(f) 25.3	(g) 2.3	(h) 1.1	(i) -24.7	(j) 26.1	(k) -26.1	(l) 26.7	(m) -27.4	(n) 27.3	(o) -29.0	(p) 28.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-11.4	-7.1	0.1	9.6	17.1	22.5	24.9	23.1	17.3	8.4	-2.2	-9.6
	DBStd	13.51	4.05	4.82	5.33	4.85	4.16	3.44	2.60	2.61	3.64	4.62	5.58	4.12
	HDD10.0	2575	662	478	311	63	2	0	0	0	2	83	367	607
	HDD18.3	4392	921	712	566	263	73	6	1	2	61	307	617	865
	CDD10.0	1775	0	0	2	53	223	374	462	406	219	35	1	0
	CDD18.3	550	0	0	0	2	36	131	203	149	28	1	0	0
	CDH23.3	5313	0	0	2	51	459	1298	1940	1247	299	18	0	0
	CDH26.7	2006	0	0	1	12	170	538	794	419	70	2	0	0
Wind	WSAvg	2.3	2.5	2.6	2.8	2.9	2.8	2.1	1.9	1.8	1.9	2.1	2.2	2.3
Precipitation	PrecAvg	386	1	1	4	10	23	71	132	92	31	16	4	1
	PrecMax	542	6	5	19	60	71	174	254	316	112	64	14	8
	PrecMin	221	0	0	0	0	2	16	35	15	3	0	0	0
	PrecStd	97	1	2	4	12	18	43	60	64	27	18	4	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.6	8.9	20.0	28.1	34.5	36.9	36.9	34.3	31.3	25.6	15.7	5.1
		MCWB	-3.2	1.4	8.0	12.6	16.2	18.2	21.8	22.1	18.1	13.1	6.1	-0.5
	2%	DB	-1.3	5.7	14.9	24.4	31.1	34.1	34.2	32.2	28.6	21.8	11.9	0.8
		MCWB	-5.2	-0.6	5.2	11.0	15.1	18.5	21.6	21.1	16.8	11.4	4.4	-3.4
	5%	DB	-3.2	3.1	11.6	21.9	28.0	31.8	32.5	30.6	26.7	18.7	8.7	-1.5
		MCWB	-6.6	-2.2	3.4	9.7	14.2	18.4	21.0	20.2	15.9	9.4	2.6	-5.0
	10%	DB	-5.2	0.6	8.9	19.1	25.5	29.8	30.8	29.0	24.8	16.2	5.9	-3.5
		MCWB	-8.1	-3.8	1.7	8.3	12.9	17.7	20.8	19.6	14.9	8.2	0.7	-6.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.2	1.7	8.0	13.7	18.0	22.5	25.6	25.1	21.0	14.1	6.4	-0.3
		MCDB	1.2	8.6	18.0	25.0	28.8	29.9	30.9	31.0	27.9	22.2	14.7	4.9
	2%	WB	-5.1	-0.5	5.5	11.6	16.2	21.2	24.4	23.5	19.1	12.3	4.4	-3.3
		MCDB	-1.4	5.4	14.4	22.4	26.8	28.2	29.9	28.6	24.7	19.8	11.3	0.6
	5%	WB	-6.6	-2.1	3.6	10.1	15.1	20.2	23.4	22.5	17.7	10.6	2.8	-5.0
		MCDB	-3.3	3.1	11.4	21.1	25.3	26.7	28.7	27.4	23.8	17.0	8.5	-1.6
	10%	WB	-8.1	-3.7	1.8	8.6	14.0	19.3	22.5	21.5	16.3	8.8	0.9	-6.4
		MCDB	-5.2	0.6	8.6	18.2	23.7	26.1	28.0	26.4	22.7	15.4	5.5	-3.5
Mean Daily Temperature Range	5% DB	MDBR	9.0	10.2	11.1	12.1	12.1	10.7	9.6	10.2	12.0	11.3	9.5	8.6
		MCDBR	11.4	13.1	15.4	16.8	16.8	14.5	12.7	12.7	15.2	15.6	13.4	11.2
		MCWBR	8.7	9.0	9.0	8.5	6.7	4.9	4.3	4.6	6.3	8.0	8.2	8.2
	5% WB	MCDBR	11.3	13.0	15.0	15.6	13.9	10.6	9.9	9.9	11.9	13.9	13.1	11.1
		MCWBR	8.6	8.9	8.9	8.4	6.1	4.4	4.5	4.8	5.9	7.8	8.1	8.2
Clear-Sky Solar Irradiance	taub	0.286	0.316	0.371	0.455	0.541	0.586	0.535	0.483	0.426	0.385	0.322	0.293	
	taud	2.272	2.209	2.094	1.907	1.754	1.760	1.942	2.050	2.116	2.121	2.217	2.226	
	Ebn at Noon	825	859	851	804	746	714	747	772	785	760	762	767	
	Edh at Noon	83	107	139	183	220	220	181	157	135	116	88	79	
All-Sky Solar Radiation	RadAvg	2.30	3.38	4.62	5.65	6.15	5.92	5.78	5.54	4.73	3.45	2.27	1.86	
	RadStd	0.14	0.20	0.26	0.38	0.37	0.47	0.44	0.25	0.26	0.23	0.18	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-0.92	N/A	N/A	N/A	N/A	N/A	+61	N/A
(47) Regional (2 neighbors)	N/A	N/A	-0.96	N/A	N/A	N/A	N/A	N/A	+85	+51

Nomenclature: See separate page